

Number theoretic algorithms

Wilson's theorem -:

→ It's one of the if-and-only-if Theorems for prime numbers

→ Theorem states that a number $p, p > 1$ is a prime if and only if

$$p \mid 1 + \lfloor (p-1) \rfloor$$

or $\lfloor p-1 \rfloor \equiv -1 \pmod{p}$ iff p is a prime

for $p=2 \Rightarrow 2$ divides $1+1$
 $= 2$

$p=3 \Rightarrow 3$ divides $1+2$
 $= 3$

for $p > 3$

if p is a composite then its positive divisors are among the integers

$$1, 2, 3, 4, \dots, p-1$$

and it's clear that $\gcd(\lfloor p-1 \rfloor, p) > 1$

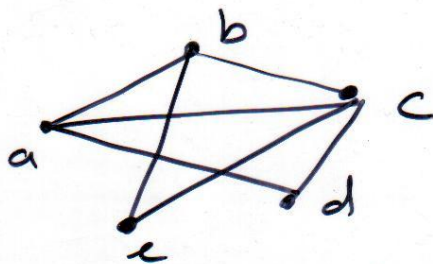
so we can not have $\lfloor p-1 \rfloor \equiv -1 \pmod{p}$.

However if p is a prime, then each of the above integers are relatively prime to p .

so for each of these integers a there is another b such that $ab \equiv 1 \pmod{p}$

$$(2 \cdot 3 \cdot 4 \dots (p-2)) \equiv 1 \pmod{p}$$

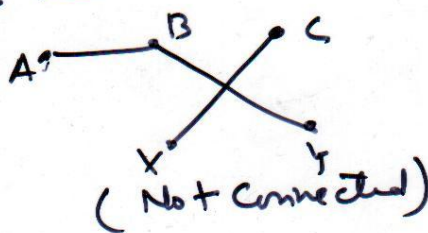
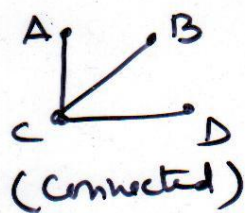
$\Rightarrow a=b$ if and only if $(a$ is 1 or $p-1$ if we omit these two other can be grouped in this way)



$$\begin{aligned} \deg(a) &= 3 & \deg(b) &= 3 & \deg(c) &= 4 \\ \deg(d) &= 2 & \deg(e) &= 2 \\ \sum \deg(u) &= 14 \Rightarrow \text{No. of edges} = 7 \end{aligned}$$

→ Connected graph :-

A graph (multigraph) G is connected if there is a path between any two of its vertices



→ Subgraph

Let G be a graph then H is a subgraph of G if $V(H) \subseteq V(G)$ vertices of H are also vertices of G and $E(H) \subseteq E(G)$ i.e. the edges of H are also edges of G .

$H(V', E')$ is a subgraph of $G(V, E)$ if $V' \subseteq V$ and $E' \subseteq E$

Primitive roots :-

→ Euler thm states that

$$\underline{a}^{\varphi(m)} \equiv 1 \pmod{m}$$

Is $\varphi(m)$ the smallest positive exponent d , of a such that $a^d \equiv 1 \pmod{m}$?

→ If it's then a is called a primitive root of m

$$\rightarrow \{ a^d \pmod{m} \}_{d=1, 2, \dots, \varphi(m)}$$

if $a=3$, $m=7$ set is

$$\{ 3, 2, 6, 4, 5, 1 \}$$

Note $d=6$, therefore 3 is primitive root of 7,

if $a=2$

$$\{ 2, 4, 1, 2, 4, 1 \}$$

We see that for this case $d=3$;
therefore 2 is not a primitive root of 7

It's not hard to see that if a is a primitive root of m then

$$\{ a, a^2, \dots, a^{\varphi(m)} \} \text{ constitutes } \varphi.$$

Reduced Residue set φ of m

→ Consists of those elements of $\mathbb{R}$ that are relatively prime to the modulus m .
ex $\mathbb{R}\mathbb{R}\mathbb{S} \pmod{10} \varphi = \{ 1, 3, 7, 9 \}$

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Quadratic Congruences and Quadratic Reciprocity Law :-

- The Quadratic residues modulo m are the integers relatively prime to m which are congruent to squares modulo m .
- integers relatively prime to m which are not congruent to square modulo m are the quadratic non residues.

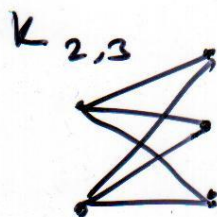
$n=17$ compute $a^2 \% n$ for each value of $a \because \gcd(a, n) = 1$
 $a \leq 1 \leq n$

- a is called Quadratic residue w.r.t b if $\exists x$ such that when a is divided by b , the remainder is same as x^2 divided by b .

Example for $n=11$
Quadratic residue modulo 11
 $= \{1, 3, 4, 5, 9\}$.

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Imp Solution



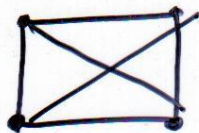
Isomorphic

suppose $G(V, E)$ and $G^*(V^*, E^*)$ are graphs and $f: V \rightarrow V^*$ is one-to-one correspondence between the set of vertices $\therefore \{u, v\}$ is an edge of G iff $\{f(u), f(v)\}$ is an edge of G^* .
 f is called isomorphism b/w G & G^* .
 G & G^* are called isomorphic graphs.



planar graph:-

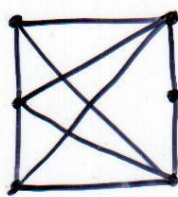
A graph or multigraph that can be drawn in a plane so that its edges do not cross is said to be planar.



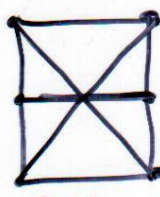
Ex



(A)
planar



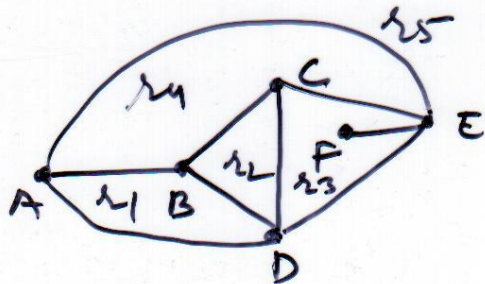
(B)
planar



(C)
Non-planar

Maps and region

→ plane representation of a finite graph is called a map.



this given map divide the plane or sphere into connected areas called regions.

$\deg(r) = \text{length of closed path bordering } r$

Thm: the sum of the degrees of the regions of a map M is equal to twice the No. of edges in M

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Euler's formulae

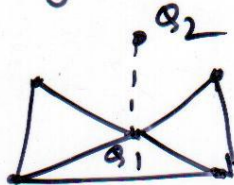
Let M be a connected map with.
 V vertices, E edges and R regions
 then $V - E + R = 2$

Proof :

P • $V = 1$
 $E = 0$
 $R = 1$

(1) $1 - 0 + 1 = 2$

otherwise M can be build up from a single vertex by the following two constructions

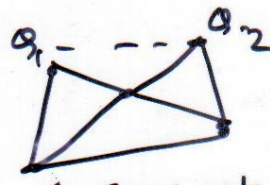


(Add a new vertex and connect it to q_1 by an edge which does not cross any existing edge)

[does not change $V - E + R$]

both incremented by 1

$V - E + R = 2$



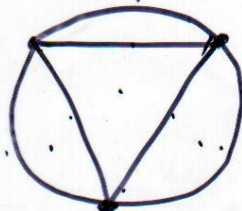
(Connect two existing vertices q_1 & q_2 by an edge which does not cross any existing edge)

[does not change $V - E + R$]

both incremented by 1.

Proof

Ex



$V = 3$
 $E = 6$
 $R = 5$

$V - E + R = 2$

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thm-: let G be a connected planar graph with p vertices and q edges
 $p \geq 3$. then

$$q \leq 3p - 6$$

$$p - q + r = 2$$

$\sum \deg(r) = 2q$
but each region has degree 3 or more

$$\therefore 2q \geq 3r$$

$$\Rightarrow r \leq \frac{2q}{3}$$

$$2 = p - q + r \leq p - q + \frac{2q}{3}$$

$$\Rightarrow \boxed{3p - 6 \geq q}$$

Colored graph:-

A coloring of a graph G is an assignment of colors to the vertices of G such that adjacent vertices have different colors.

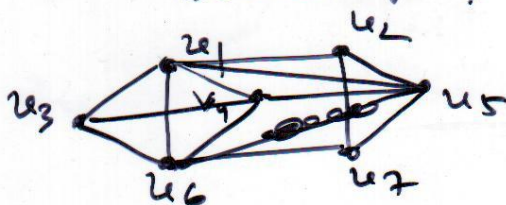
G is n -colorable if $\exists$ a coloring of G which uses n color.

The minimum No. of colors needed to paint the graph G is called the Chromatic Number ($\chi(G)$)
 χ

Welch - Powell algorithm for painting a graph G

- order the vertices of G in decreasing degree
- Use one color to paint the first vertex and to paint in sequential order, each vertex on the list that is not adjacent to a vertex previously painted with the same color.
- start again at the top of the list and repeat the process painting previously unpainted vertices using a second color.
- Continue repeating with additional colors until all the vertices have been painted

Note: - This also only give upper bound of $\chi(G)$.

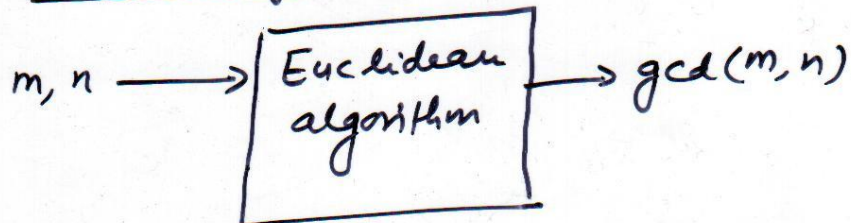


	u_1	u_4	u_5	u_6	u_2	u_3	u_7	
degree	5	4	4	4	3	3	3	
color	a	b	c	c	b	d	a	

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Number theoretic algorithm

* → Euclid algorithm to compute GCD



Euclid (m, n)

```
{  
   $x = m, y = n$   
  while ( $y > 0$ )  
     $r = x \bmod y$   
     $x = y$   
     $y = r$ 
```

return x

}

Example: $\text{gcd}(33, 77)$

$x = 33 \quad y = 77$

→ $x = 77$
 $y = 33$

→ $x = 33$
 $y = 11$

→ $x = 11$ $\leftarrow \text{gcd}(33, 77)$
 $y = 0$

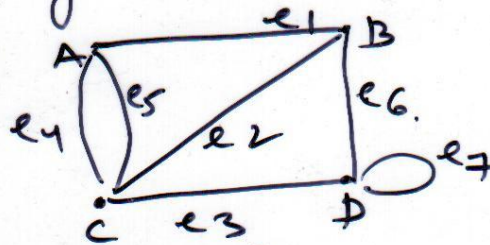
Graphs (Graph Terminology)

graph algorithms

→ Graph:- $G \equiv (V, E)$

→ Multigraph:-

A multigraph $G = G(V, E)$ also consists of a set V of vertices and a set E of edges except that E may contain multiple edges i.e. edges connecting the same endpoints and E may contain one or more loops.



→ Trivial graph

→ The trivial graph is the graph with one vertex and no edges

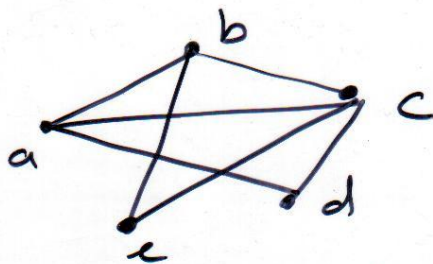
→ The empty graph is the graph with no vertices and no edges.

→ Degree and parity (even or odd)

of a vertex:- The degree of a vertex v in a graph G is $\deg(v) = \text{No. of edges which are incident on } v$

Thm: sum of the degree of the vertices = twice the Number of edges

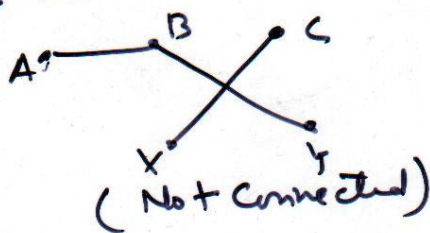
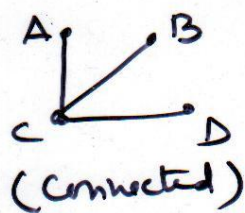
(1)



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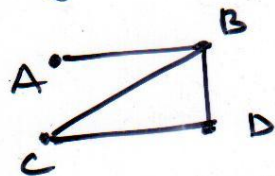


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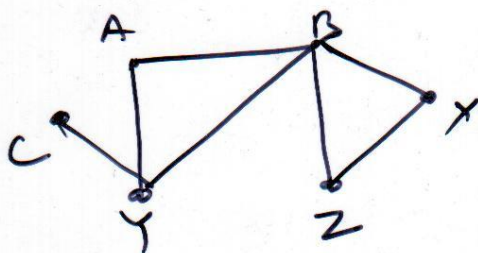
$H(V', E')$ is a subgraph of $G(V, E)$ if $V' \subseteq V$ and $E' \subseteq E$

Suppose $H = H(V', E')$ is a subgraph of $G = G(V, E)$ then H is called a full subgraph of G if E' contains all the edges of E whose endpoints lie in V' .



$V' = \{A, B, C\}$ $E' = \{AB, AC\}$ Not a subgraph.
 $V' = \{A, B, C\}$ and $E' = \{AB, AC, BC\}$ Yes

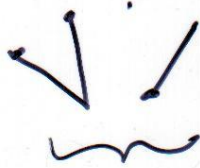
cut-point : A vertex v is called a cutpoint for G if $G - v$ is disconnected



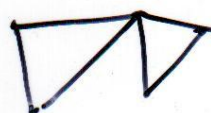
$G - A$



$G - B$



$G - C$



B is cutpoint

Complete graph:-

A Graph G is complete if each vertex is connected to every other vertex. (K_n) -

K_1

.

K_2

—

K_3



K_4



K_5



No. of edges
 $\frac{n(n-1)}{2}$

Regular graph:-

A graph G is regular of degree k or k -regular if every vertex has degree k . In other words, a graph is regular if every vertex has the same degree.



2 regular

bipartite and complete bipartite graph.

A graph G is said to be bipartite if its vertices V can be partitioned into two subsets M and N such that each edge of G connects a vertex of M to a vertex of N .

complete bipartite = each vertex of M is connected to each vertex of N . $K_{m,n}$

(4)